7(a)(i)	kinetic energy (of the fluid)	B1
7(a)(ii)	gravitational potential energy (of the fluid)	B1
7(b)(i)	They are the same/ equal.	B1
7(b)(ii)	mass flow (rate) at inlet = mass flow (rate) at outlet	B1

$$\frac{m_A}{t} = \frac{m_B}{t}$$

$$\frac{\rho \frac{\pi {D_A}^2}{4} l_A}{t} = \frac{\rho \frac{\pi {D_B}^2}{4} l_B}{t}$$

B1

$${D_A}^2 v_A = {D_B}^2 v_B$$

$$\frac{v_A}{v_B} = \frac{{D_B}^2}{{D_A}^2}$$

Question		Marks
7(b)(iii)	$\frac{v_{out}}{v_{in}} = \frac{D_{in}^{2}}{D_{out}^{2}}$	
	$v_{out} = \left(\frac{0.100}{0.050}\right)^2 v_{in}$	
	$= 4v_{in}$	C1
	$\frac{P_{in}}{\rho g} + \frac{v_{in}^{2}}{2g} + h_{in} = \frac{P_{out}}{\rho g} + \frac{v_{out}^{2}}{2g} + h_{out}$	M1
	$\frac{P_{out} - P_{in}}{\rho g} = \frac{v_{in}^{2} - v_{out}^{2}}{2g}$	
	$\Delta P = \frac{\rho\left(v_{in}^{2} - v_{out}^{2}\right)}{2}$	
	$= \frac{\rho\left(v_{in}^{2} - \left(4v_{in}\right)^{2}\right)}{2}$	
	$= -480\ \text{kPa}$	A1
7(c)(i)	The pressure difference will be higher.	B1
7(c)(ii)	$v_{ave} = 2.5v_{in}$ $D_{ave} = 0.075\ \text{m}$ (or any other appropriate averaging done)	C1
	$p_{Loss} = H_{Loss} \times \rho g$	
	$= f_D \frac{v_{ave}^{2} L}{2gD_{ave}} \times \rho g$	
	$= 0.0038 \times \frac{(2.5 \times 8)^2 \times 10}{2 \times 0.075} \times 1000$	A1
	$= 101\ \text{kPa}$	
7(d)(i)	✓ Blood vessels are flexible ✓ (Higher) friction factor / viscosity of blood ✓ Blood is a suspension rather than a pure fluid. ✓ Capillary action due to the small diameter of blood vessels.	B1
7(d)(ii)	As the arteries narrow, the model predicts that the <u>velocity of the flow</u> at this cross section will be <u>higher</u>.	M1
	From the expression in (a), the <u>pressure of the blood</u> at the narrow cross-section will be <u>lower</u>.	M1
	Thus, this model proposed by the student <u>will not be able to correctly predict</u> this observation.	A1

Question		Marks
8(c)	gravitational force provides (just enough) centripetal acceleration of	